

EDA Interview Questions & Answers

Beginner → Advanced | Data Analyst / Data Scientist / ML Internship

Quick Definition

EDA (Exploratory Data Analysis) is the process of examining, understanding, cleaning, and analyzing a dataset before building a Machine Learning or statistical model.

1. Basic EDA Questions

Q1. What is EDA?

Answer: EDA stands for Exploratory Data Analysis. It is used to understand the structure, quality, patterns, relationships, and distribution of data before modeling.

Q2. Why do we perform EDA?

Answer: To understand the dataset, find missing values and outliers, discover patterns, detect errors, and decide what preprocessing and feature engineering are needed.

Q3. What are the main steps of EDA?

Answer: Understand the dataset → inspect data types → check missing/duplicate values → analyze distributions → detect outliers → study relationships → visualize findings → prepare data for modeling.

Q4. Which Python libraries are commonly used for EDA?

Answer: Pandas and NumPy are commonly used for data manipulation and statistics. Matplotlib and Seaborn are widely used for visualization. Other automated EDA tools include Sweetviz and ydata-profiling.

Q5. What is the difference between EDA and Data Cleaning?

Answer: Data cleaning focuses on fixing data-quality problems such as missing, duplicate, or invalid values. EDA is broader: it explores the data and helps us understand what cleaning and transformations are appropriate.

Q6. What are univariate, bivariate, and multivariate analysis?

Answer: Univariate analyzes one variable, bivariate analyzes two variables and their relationship, and multivariate analyzes three or more variables together.

Q7. What is the purpose of `df.info()`?

Answer: It shows the number of rows, columns, non-null counts, and data types. It is useful for quickly checking dataset structure and missing values.

Q8. What is the purpose of `df.describe()`?

Answer: It provides descriptive statistics for numerical columns, such as count, mean, standard deviation, minimum, quartiles, and maximum.

Q9. How do you check missing values in Pandas?

Answer: Use `df.isnull().sum()` to get the number of missing values in each column.

Q10. How do you identify duplicate records?

Answer: Use `df.duplicated().sum()` to count duplicate rows and `df[df.duplicated()]` to inspect them.

2. Missing Values

Q1. How do you handle missing values?

Answer: Depending on the situation, you can remove rows/columns, fill numerical values with mean or median, fill categorical values with mode, or use more advanced imputation methods.

Q2. When would you use mean, median, or mode?

Answer: Mean is useful when data is reasonably symmetric. Median is often better when data is skewed or contains outliers. Mode is commonly used for categorical values.

Q3. Why is median sometimes preferred over mean?

Answer: Median is less affected by extreme values, so it is more robust for skewed data and datasets containing outliers.

Q4. When should you drop missing values?

Answer: Drop them when the missing portion is small and the remaining observations are sufficient, or when a column has very high missingness and is not important. Always consider the business context before dropping data.

3. Outliers

Q1. What is an outlier?

Answer: An outlier is an observation that is unusually far from the general pattern of the data.

Q2. How do you detect outliers?

Answer: Common methods include boxplots, the IQR method, Z-scores, scatter plots, and domain-specific rules.

Q3. What is the IQR method?

Answer: $IQR = Q3 - Q1$. Values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ are commonly flagged as potential outliers.

Q4. What is a boxplot?

Answer: A boxplot summarizes a numerical distribution using quartiles and helps visualize spread and potential outliers.

Q5. Should we always remove outliers?

Answer: No. Some outliers are valid and meaningful. Remove or transform them only when there is evidence that they are errors or harmful for the specific analysis/model.

4. Distribution & Statistics

Q1. What is skewness?

Answer: Skewness measures the asymmetry of a distribution. Positive skew generally has a longer right tail, while negative skew generally has a longer left tail.

Q2. What is kurtosis?

Answer: Kurtosis describes the tail heaviness and concentration of a distribution relative to a normal distribution.

Q3. What is a normal distribution?

Answer: A symmetric, bell-shaped probability distribution where mean, median, and mode are equal at the center.

Q4. Why is understanding distribution important?

Answer: It helps select appropriate transformations, identify outliers, understand model assumptions, and choose suitable statistical techniques.

Q5. Mean vs median — when would you use each?

Answer: Use mean when the data is reasonably symmetric without strong outliers. Prefer median when the distribution is skewed or contains extreme values.

5. Visualization

Q1. When would you use a histogram?

Answer: To understand the distribution, spread, center, and shape of a numerical variable.

Q2. When would you use a boxplot?

Answer: To compare distributions across groups and quickly identify spread and potential outliers.

Q3. When would you use a scatter plot?

Answer: To visualize the relationship between two numerical variables and identify trends, clusters, or unusual observations.

Q4. Bar chart vs histogram?

Answer: A bar chart is typically used for categorical values and has separated bars. A histogram is used for numerical data grouped into continuous intervals, so the bars normally touch.

Q5. When would you use a heatmap?

Answer: A heatmap can show correlation matrices or other matrix-like values, making strong and weak relationships easier to spot.

6. Correlation & Relationships

Q1. What is correlation?

Answer: Correlation measures the strength and direction of association between two variables. Pearson correlation ranges from -1 to +1.

Q2. What do +1, 0, and -1 mean?

Answer: +1 indicates perfect positive linear correlation, 0 indicates no linear correlation, and -1 indicates perfect negative linear correlation.

Q3. Does correlation imply causation?

Answer: No. Two variables can be correlated because of coincidence, a third variable, reverse causality, or other factors.

Q4. How do you find highly correlated features?

Answer: Use a correlation matrix, for example `df.corr(numeric_only=True)`, and inspect feature pairs with very high absolute correlation.

Q5. What is multicollinearity?

Answer: Multicollinearity occurs when two or more predictor variables are strongly related. It can make coefficient-based models unstable and harder to interpret.

7. Practical Interview Questions

Q1. You receive a new dataset. What will you do first?

Answer: First understand the business problem and target variable. Then inspect shape, columns, data types, sample rows, missing values, duplicates, distributions, outliers, and relationships before deciding on preprocessing.

Q2. How would you perform EDA from scratch?

Answer: A practical workflow is: load data → understand the problem → inspect shape and schema → clean obvious issues → analyze missing values and duplicates → study distributions → detect outliers → analyze relationships → summarize findings → plan preprocessing/features.

Q3. How do you identify columns that need preprocessing?

Answer: Check data types, missingness, unique values, inconsistent categories, scale, distributions, and outliers. This helps decide on encoding, imputation, scaling, or transformations.

Q4. How do you handle categorical variables?

Answer: For low-cardinality nominal variables, one-hot encoding is common. Ordinal variables can use an order-aware encoding. The choice depends on the meaning and model.

Q5. How do you identify irrelevant features?

Answer: Use domain knowledge, unique-value checks, correlation analysis, feature importance, statistical tests where appropriate, and model-based feature selection. Avoid using the target or future information incorrectly.

Q6. What is data leakage and how can EDA help detect it?

Answer: Data leakage occurs when information unavailable at prediction time enters the training process. EDA can reveal suspicious features, future timestamps, target-derived columns, or unusually strong relationships that need investigation.

Q7. What if one column has 50% missing values?

Answer: Do not automatically drop it. First determine why values are missing and whether the column is important. Consider imputation, a missing-indicator feature, redesigning the feature, or dropping it if justified.

Q8. What if a feature has extreme outliers?

Answer: Investigate whether they are errors or valid observations. Depending on the context, you may correct errors, cap/winsorize, transform the feature, use robust methods, or retain the values.

Q9. How do you find relationships between numerical and categorical variables?

Answer: You can compare numerical distributions across categories using boxplots/violin plots and summary statistics. Statistical tests may also be appropriate depending on assumptions.

Q10. After EDA, how do you decide which features to use?

Answer: Combine domain knowledge, data quality, leakage checks, redundancy/correlation analysis, feature engineering, and validation-based feature selection. Final decisions should be tested using a proper train/validation setup.

Useful Python EDA Commands

Purpose	Python
View first rows	<code>df.head()</code>
Dataset shape	<code>df.shape</code>
Column information	<code>df.info()</code>
Summary statistics	<code>df.describe()</code>
Missing values	<code>df.isnull().sum()</code>
Duplicate rows	<code>df.duplicated().sum()</code>
Unique values	<code>df['column'].nunique()</code>
Value counts	<code>df['column'].value_counts()</code>
Correlation	<code>df.corr(numeric_only=True)</code>

Interview Tip

Do not memorize only definitions. Be ready to explain **why** you used a particular EDA technique and what decision you made from the result. For example: "I used a boxplot to identify potential outliers, then investigated whether those values were genuine before deciding whether to keep or transform them."